

Questions with Answer Keys

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Q1 - Properties of Definite Integrals - I

The value of  $\int_0^2 |x^2 + 2x - 3| dx$  is

- (1) 1
- (2) 2
- (3) 3
- (4) 4

Q2 - Properties of Definite Integrals - I

The value of  $\int_0^2 [x^2] dx$  (where,  $[\cdot]$  denotes the greatest integer function) is

- (1)  $\sqrt{3} - \sqrt{2}$
- (2)  $5 - \sqrt{3} - \sqrt{2}$
- (3)  $\sqrt{5} - \sqrt{3}$
- (4) None of these

Q3 - Properties of Definite Integrals - I

$\int_0^a \{x^n\} dx$  (where,  $\{\cdot\}$  denotes fractional part of  $x$ ) is

- (1)  $\sum_{r=1}^k (r-1) \{r^{1/n} - (r-1)^{1/n}\} + k(a - k^{1/n})$
- (2)  $\sum_{r=1}^k (r-1) \{r^{1/n} - (r-1)^{1/n}\}$
- (3)  $\sum_{r=1}^k \{r^{1/n} - (r-1)^{1/n}\} + k(a - k^{1/n})$
- (4) None of the above

Q4 - Properties of Definite Integrals - I

$\int_0^2 [x^2 - x + 1] dx$  (where,  $[\cdot]$  denotes the greatest integer function) is equal to

- (1)  $\frac{5-\sqrt{5}}{2}$
- (2)  $\frac{5+\sqrt{5}}{2}$
- (3)  $\frac{1+\sqrt{5}}{2}$
- (4)  $\frac{\sqrt{5}-1}{2}$

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## Q5 - Properties of Definite Integrals - I

$\int_0^{100} [\tan^{-1} x] dx$ , where  $[\cdot]$  denotes the greatest integer function, is equal to

- (1) 1000
- (2)  $-\tan 1$
- (3) 0
- (4)  $100 - \tan 1$

## Q6 - Properties of Definite Integrals - I

$\int_{1/\theta}^{\tan x} \frac{t}{1+t^2} dt + \int_{1/\theta}^{\cot x} \frac{1}{t(1+t^2)} dt$  is equal to

- (1)  $e$
- (2)  $1/e$
- (3) 1
- (4)  $e - 1$

## Q7 - Properties of Definite Integrals - I

The value of  $\int_1^a [x] f'(x) dx$ ,  $a > 1$  (where,  $[\cdot]$  denotes the greatest integer function), is

- (1)  $[a] f(a) - \{f(1) + f(2) + \dots + f([a])\}$
- (2)  $[a] f([a]) - \{f(1) + f(2) + \dots + f(a)\}$
- (3)  $a f([a]) - \{f(1) + f(2) + \dots + f([a])\}$
- (4)  $a f(a) - \{f(1) + f(2) + \dots + f([a])\}$

## Q8 - Properties of Definite Integrals - II

Let  $f(x)$  be a function satisfying  $f'(x) = f(x)$  with  $f(0) = 1$  and  $g(x)$  be the function satisfying  $f(x) + g(x) = x^2$ . Then, the value of  $\int_0^1 f(x)g(x)dx$  is

- (1)  $e - \frac{1}{2}e^2 - \frac{5}{2}$
- (2)  $e - e^2 - 3$
- (3)  $\frac{1}{2}(e - 3)$

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(4)  $e - \frac{e^2}{2} - \frac{3}{2}$

## Q9 - Properties of Definite Integrals - II

For  $f(x) = x^4 + |x|$ , let  $I_1 = \int_0^\pi f(\cos x)dx$  and  $I_2 = \int_0^{\pi/2} f(\sin x)dx$ , then  $\frac{I_1}{I_2}$  has the value

(1) 1

(2)  $1/2$

(3) 2

(4) 4

## Q10 - Properties of Definite Integrals - II

$\int_0^1 x(1-x)^n dx$  is equal to

(1)  $\frac{1}{n+1} + \frac{1}{n+2}$

(2)  $\frac{1}{n+1}$

(3)  $\frac{1}{n+2}$

(4)  $\frac{1}{n+1} - \frac{1}{n+2}$

## Q11 - Properties of Definite Integrals - II

If  $\int_0^\pi x f(\sin x) dx = A \int_0^{\pi/2} f(\sin x) dx$ , then the value of  $A$  is

(1) 0

(2)  $\pi$

(3)  $\pi/4$

(4)  $2\pi$

## Q12 - Properties of Definite Integrals - II

$\int_{-3\pi/2}^{\pi/2} \{(x+\pi)^3 + \cos^2(x+3\pi)\} dx$  is equal to

(1)  $\frac{\pi^4}{32} + \frac{\pi}{2}$

(2)  $\frac{\pi}{2}$

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(3)  $\frac{\pi}{4} - 1$

(4)  $\frac{\pi^4}{32}$

Q13 - Properties of Definite Integrals - II

The value of  $\int_0^\pi [\cot x] dx$  (where,  $[\cdot]$  denotes the greatest integer function) is

(1)  $-1$

(2)  $-\pi/2$

(3)  $\pi/2$

(4)  $1$

Q14 - Properties of Definite Integrals - II

$\int_0^\pi e^{\cos^2 x} \cos^3(2n+1)x dx$  is equal to

(1)  $\pi$

(2)  $1$

(3)  $0$

(4) None of these

Q15 - Properties of Definite Integrals - II

$\int_{-1/2}^{1/2} \left[ [x] + \log\left(\frac{1+x}{1-x}\right) \right] dx$  is equal to

(1)  $-1/2$

(2)  $0$

(3)  $1$

(4)  $2 \log(1/2)$

Q16 - Properties of Definite Integrals - II

$\int_{-\pi/2}^{\pi/2} \left[ x^2 + \log\left(\frac{\pi+x}{\pi-x}\right) \right] \cos x dx$  is

(1)  $0$

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(2)  $\frac{\pi^2}{2} - 4$

(3)  $\frac{\pi^2}{2} + 4$

(4)  $\frac{\pi^2}{2}$

## Q17 - Periodic Function and Leibnitz Theorem

Let  $f : R \rightarrow R$  be a differentiable function and  $f(1) = 4$ , then the value of  $\lim_{x \rightarrow 1} \int_4^{f(x)} \frac{2t}{x-1} dt$  is

(1)  $8f'(1)$

(2)  $4f'(1)$

(3)  $2f'(1)$

(4)  $f'(1)$

## Q18 - Periodic Function and Leibnitz Theorem

For  $n \in N$ , the value of  $\int_0^{n\pi+v} \sqrt{\frac{1+\cos 2x}{2}} dx$ , where  $\frac{\pi}{2} < v < \pi$  is

(1)  $2n + 1 - \cos v$

(2)  $2n - \sin v$

(3)  $2n + 2 - \sin v$

(4)  $2n + 1 - \sin v$

## Q19 - Periodic Function and Leibnitz Theorem

If  $f(x) = \int_0^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_0^{\cos^2 x} \cos^{-1} \sqrt{t} dt$ ,  $x \in [0, \pi/2]$ , then  $f(x)$  equals

(1)  $\pi/2$

(2)  $\pi/4$

(3)  $\pi$

(4)  $0$

## Q20 - Periodic Function and Leibnitz Theorem

Let  $f : R \rightarrow R$  be a differentiable function having  $f(2) = 6$ ,  $f'(2) = \frac{1}{48}$ . Then,  $\lim_{x \rightarrow 2} \int_6^{f(x)} \frac{4t^3}{x-2} dt$  equals

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(1) 24

(2) 36

(3) 12

(4) 18

## Q21 - Periodic Function and Leibnitz Theorem

Let  $f : (0, \infty) \rightarrow R$  and  $F(x) = \int_0^x f(t)dt$ . If  $F(x^2) = x^2(1+x)$ , then  $f(4)$  is equal to

(1)  $5/4$ 

(2) 7

(3) 4

(4) 2

## Q22 - Periodic Function and Leibnitz Theorem

Let  $T > 0$  be a fixed real number. Suppose  $f$  is a continuous function such that for all

$x \in R, f(x+T) = f(x)$ . If  $I = \int_0^T f(x)dx$ , then the value of  $\int_3^{3+3T} f(2x)dx$  is

(1)  $\frac{3}{2} I$ (2)  $2 I$ (3)  $3 I$ (4)  $6 I$ 

## Q23 - Periodic Function and Leibnitz Theorem

$f(x) = \int_{x^2}^{x^2+1} e^{-t^2} dt$  increases in the interval

(1) no where

(2)  $x \leq 0$ (3)  $x \in [-2, 2]$ (4)  $x \geq 0$

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## Q24 - Periodic Function and Leibnitz Theorem

Let  $f$  be a non-negative function defined on the interval  $[0, 1]$ . If  $\int_0^x \sqrt{1 - [f'(t)]^2} dt = \int_0^x f(t) dt$ ,  $0 \leq x \leq 1$  and  $f(0) = 0$ , then

(1)  $f(1/2) < 1/2$  and  $f(1/3) > 1/3$

(2)  $f(1/2) > 1/2$  and  $f(1/3) > 1/3$

(3)  $f(1/2) < 1/2$  and  $f(1/3) < 1/3$

(4)  $f(1/2) > 1/2$  and  $f(1/3) < 1/3$

## Q25 - Periodic Function and Leibnitz Theorem

Let  $f$  be a real-valued function defined on the interval  $(-1, 1)$  such that  $e^{-x} f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt$ , for all  $x \in (-1, 1)$  and let  $f^{-1}$  be the inverse function of  $f$ . Then,  $(f^{-1})'(2)$  is equal to

(1) 1

(2)  $1/3$

(3)  $1/2$

(4)  $1/e$

## Q26 - Estimation and Gamma Function

Let  $I = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$  and  $J = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$ , then which one of the following is true?

(1)  $I > 2/3$  and  $J > 2$

(2)  $I < 2/3$  and  $J < 2$

(3)  $I < 2/3$  and  $J > 2$

(4)  $I > 2/3$  and  $J < 2$

## Q27 - Estimation and Gamma Function

If  $l(m, n) = \int_0^1 t^m (1+t)^n dt$ , then  $l(m, n)$  in terms of  $l(m+1, n-1)$  is

(1)  $\frac{m}{n+1} l(m+1, n-1)$

(2)  $\frac{n}{m+1} l(m+1, n-1)$

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(3)  $\frac{2^n}{m+1} - \frac{n}{m+1} l(m+1, n-1)$

(4)  $\frac{2^n}{m+1} - \frac{n}{n+1} l(m+1, n-1)$

Q28 - Estimation and Gamma Function

The absolute value  $\int_{10}^{19} \frac{(\sin x)}{1+x^8} dx$  is less than

(1)  $10^{-10}$

(2)  $10^{-11}$

(3)  $10^{-7}$

(4)  $10^{-9}$

Q29 - Limit as a Sum

Let  $S_n = \frac{n}{(n+1)(n+2)} + \frac{n}{(n+2)(n+4)} + \frac{n}{(n+3)(n+6)} + \dots + \frac{1}{(6n)}$  then  $\lim_{n \rightarrow \infty} S_n$  is equal to

(1)  $\log\left(\frac{3}{2}\right)$

(2)  $\log\left(\frac{9}{2}\right)$

(3) 1

(4) 0

Q30 - Limit as a Sum

The value of  $\lim_{n \rightarrow \infty} \frac{1^p + 2^p + \dots + n^p}{n^{p+1}}$  is

(1) 1

(2)  $\frac{1}{p+1}$

(3)  $\frac{1}{p+2}$

(4)  $p^2$

Q31 - Limit as a Sum

The value of  $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} e^{r/n}$  is

(1)  $e$

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(2)  $e - 1$

(3)  $1 - e$

(4)  $e + 1$

**Q32 - Limit as a Sum**

The value of  $\lim_{n \rightarrow \infty} \left( \frac{1}{n^2} \sec^2 \frac{1}{n^2} + \frac{2}{n^2} \sec^2 \frac{4}{n^2} + \cdots + \frac{1}{n} \sec^2 1 \right)$  is

(1)  $\frac{1}{2} \sec 1$

(2)  $\frac{1}{2} \operatorname{cosec} 1$

(3)  $\tan 1$

(4)  $\frac{1}{2} \tan 1$

**Q33 - Limit as a Sum**

The value of  $\lim_{n \rightarrow \infty} \sum_{r=1}^{4n} \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r}+4\sqrt{n})^2}$  is

(1)  $1/35$

(2)  $1/14$

(3)  $1/10$

(4)  $1/5$

**Q34 - Area under Curves**

If the area bounded by the  $X$ -axis, curve  $y = f(x)$  and the lines  $x = 1, x = b$  is equal to  $\sqrt{b^2 + 1} - \sqrt{2}$  for all  $b > 1$ , then  $f(x)$  is equal to

(1)  $\sqrt{x+1}$

(2)  $\sqrt{x-1}$

(3)  $\sqrt{x^2+1}$

(4)  $x/\sqrt{1+x^2}$

**Q35 - Area under Curves**

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The area of the region bounded by the curves  $y = |x - 1|$  and  $y = 3 - |x|$ , is

(1) 6 sq units

(2) 2 sq units

(3) 3 sq units

(4) 4 sq units

## Q36 - Area under Curves

The parabolas  $y^2 = 4x$  and  $x^2 = 4y$  divide the square region bounded by  $x = 4$ ,  $y = 4$  and the coordinate axes. If  $S_1$ ,  $S_2$  and  $S_3$  are respectively the area of these parts numbered from top to bottom, then  $S_1 : S_2 : S_3$  is

(1) 1 : 2 : 1

(2) 1 : 2 : 3

(3) 2 : 1 : 2

(4) 1 : 1 : 1

## Q37 - Area under Curves

The area of the plane region bounded by the curves  $x + 2y^2 = 0$  and  $x + 3y^2 = 1$ , is

(1)  $\frac{4}{3}$  sq units

(2)  $\frac{5}{3}$  sq units

(3)  $\frac{2}{3}$  sq units

(4)  $\frac{1}{3}$  sq units

## Q38 - Area under Curves

The area (in sq units) bounded by the curves  $y = \cos x$ ,  $y = \sin x$  between the ordinates  $x = 0$  and  $x = 3\pi/2$ , is

(1)  $4\sqrt{2} - 2$

(2)  $4\sqrt{2} + 2$

(3)  $4\sqrt{2} - 1$

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(4)  $4\sqrt{2} + 1$

## Q39 - Area under Curves

The area (in square units) of region bounded by the curves  $y = x^2$ ,  $y = |2 - x^2|$  and  $y = 2$ , which lies to the right of line  $x = 1$ , is

(1)  $\frac{20}{3} - 4\sqrt{2}$

(2)  $\frac{20}{3} + 4\sqrt{2}$

(3)  $\frac{10}{3} - 4\sqrt{2}$

(4)  $\frac{10}{3} - 2\sqrt{2}$

## Q40 - Area under Curves

The area enclosed between the curves  $y = ax^2$  and  $x = ay^2$  ( $a > 0$ ) is 1 sq unit, then the value of  $a$  is

(1)  $1/\sqrt{3}$

(2)  $1/2$

(3) 1

(4)  $1/3$

## Q41 - Area under Curves

Let  $A(m)$  be area bounded by parabola  $y = x^2 + 2x - 3$  and the line  $y = mx + 1$ , then the least area of  $A(m)$  is

(1)  $\frac{8}{3}$  units

(2)  $\frac{16}{3}$  sq units

(3)  $\frac{32}{3}$  sq units

(4) None of these

## Q42 - Area under Curves

The area enclosed by the curves  $y = 2x^2$  and  $y = x^4 - 2x^2$  is

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(1)  $\frac{128}{11}$  sq units

(2)  $\frac{128}{15}$  sq units

(3) 128 sq units

(4) None of these

Q43 - Area under Curves

The area (in sq units) of region bounded by the curves  $y = x^2$  and  $y = \sec^{-1}[-\sin^2 x]$  where  $[\cdot]$  denotes the greatest integer function, is

(1)  $4\pi\frac{\sqrt{\pi}}{3}$

(2)  $\frac{4\sqrt{\pi}}{3}$

(3)  $\frac{4\pi^2}{3}$

(4) None of these

Q44 - Area under Curves

Suppose  $y = f(x)$  and  $y = g(x)$  are two functions whose graphs intersect at three points  $(0, 4)$ ,  $(2, 2)$  and  $(4, 0)$  with  $f(x) > g(x)$  for  $0 < x < 2$  and  $f(x) < g(x)$  for  $2 < x < 4$ . If  $\int_0^4 [f(x) - g(x)]dx = 10$  and  $\int_2^4 [g(x) - f(x)]dx = 5$ . The area between two curves for  $0 < x < 2$  is

(1) 5 sq units

(2) 10 sq units

(3) 15 sq units

(4) 20 sq units

Q45 - Area under Curves

Let  $a$  be a positive constant number. Consider two curves  $C_1 : y = e^x$ ,  $C_2 : y = e^{a-x}$ . Let  $S$  be the area of the part surrounding by  $C_1$ ,  $C_2$  and  $Y$ -axis, then  $\lim_{a \rightarrow 0} \frac{S}{a^2}$  equals

(1) 4

(2)  $\frac{1}{2}$

## JEE Advanced Crash Course

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Answer Key

|         |         |         |         |
|---------|---------|---------|---------|
| Q1 (4)  | Q2 (2)  | Q3 (1)  | Q4 (1)  |
| Q5 (4)  | Q6 (3)  | Q7 (1)  | Q8 (4)  |
| Q9 (3)  | Q10 (4) | Q11 (2) | Q12 (2) |
| Q13 (2) | Q14 (3) | Q15 (1) | Q16 (2) |
| Q17 (1) | Q18 (2) | Q19 (2) | Q20 (4) |
| Q21 (3) | Q22 (3) | Q23 (2) | Q24 (3) |
| Q25 (2) | Q26 (2) | Q27 (3) | Q28 (3) |
| Q29 (1) | Q30 (2) | Q31 (2) | Q32 (4) |
| Q33 (3) | Q34 (4) | Q35 (4) | Q36 (4) |
| Q37 (1) | Q38 (1) | Q39 (1) | Q40 (1) |
| Q41 (3) | Q42 (2) | Q43 (1) | Q44 (3) |
| Q45 (4) |         |         |         |